



National Textile University

Department of Computer Science

Subject:

Operating System

Submitted to:

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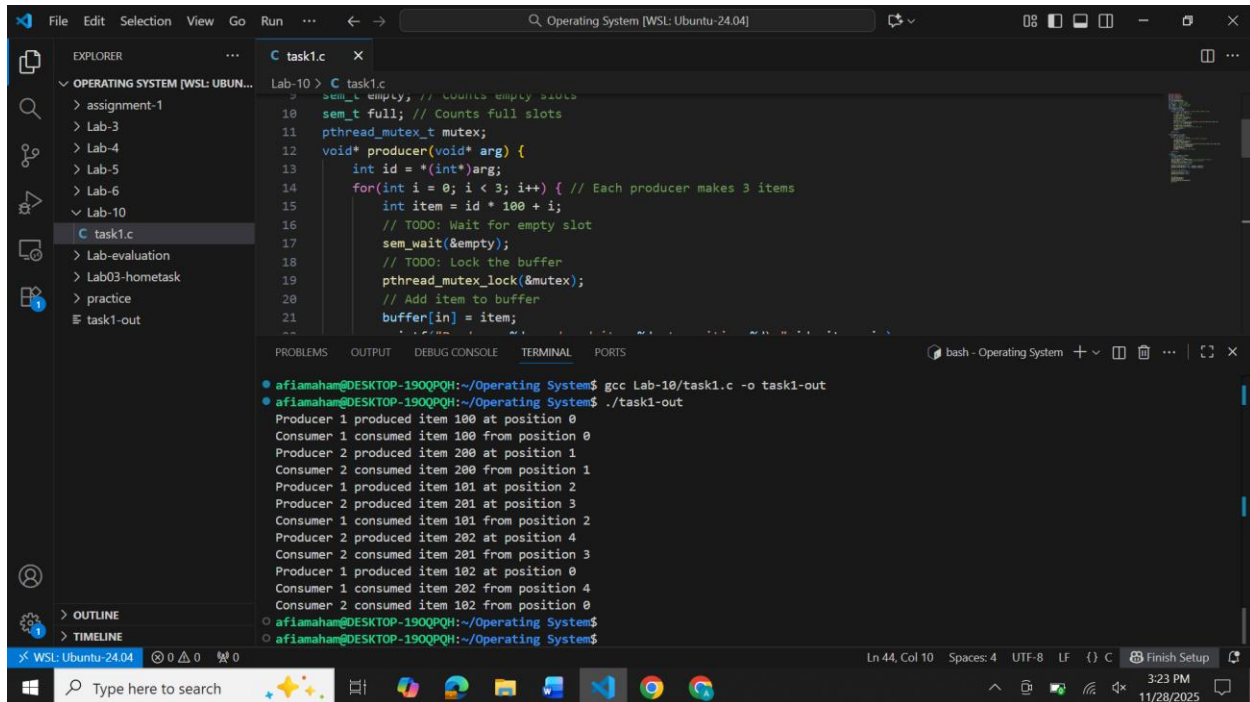
Lab no:

10

Semester:

5th

Task 01:



The screenshot shows a Visual Studio Code editor window with the file explorer on the left displaying a project structure under 'OPERATING SYSTEM [WSL: Ubuntu-24.04]'. The file 'task1.c' is selected. The code in 'task1.c' is a C program for a producer-consumer problem. It defines a semaphore 'sem_t' with 'empty' and 'full' slots, a mutex, and a buffer. The 'producer' function generates items and adds them to the buffer. The 'consumer' function removes items from the buffer. The main function creates two producers and two consumers, each running in its own thread. The terminal at the bottom shows the compilation and execution of the program, resulting in a sequence of producer and consumer actions and item values.

```
Lab-10 > C task1.c
10 sem_t empty; // Counts empty slots
11 sem_t full; // Counts full slots
12 pthread_mutex_t mutex;
13 void* producer(void* arg) {
14     int id = *(int*)arg;
15     for(int i = 0; i < 3; i++) { // Each producer makes 3 items
16         int item = id * 100 + i;
17         // TODO: Wait for empty slot
18         sem_wait(&empty);
19         // TODO: Lock the buffer
20         pthread_mutex_lock(&mutex);
21         // Add item to buffer
22         buffer[in] = item;
23     }
24 }
```

```
afiamaham@DESKTOP-190QPQH:~/Operating System$ gcc Lab-10/task1.c -o task1-out
afiamaham@DESKTOP-190QPQH:~/Operating System$ ./task1-out
Producer 1 produced item 100 at position 0
Consumer 1 consumed item 100 from position 0
Producer 2 produced item 200 at position 1
Consumer 2 consumed item 200 from position 1
Producer 1 produced item 101 at position 2
Producer 2 produced item 201 at position 3
Consumer 1 consumed item 101 from position 2
Producer 2 produced item 202 at position 4
Consumer 2 consumed item 201 from position 3
Producer 1 produced item 102 at position 0
Consumer 1 consumed item 202 from position 4
Consumer 2 consumed item 102 from position 0
afiamaham@DESKTOP-190QPQH:~/Operating System$
```

Task 02:

How many threads are present in the code?

Total 4 threads are present. 2 producers and 2 consumers.

Task 03:

What output following line will produce?

```
int item = id * 100 + i;
```

- For producer of id 1, the output will be 100, 101, 102 (As each producer will produce 3 values)
- For producer of id 2, the output will be 200, 201, 202

Task 04:

What happens if producer is producing 3 values while consumer is consuming 4 values?

Consumer will wait forever and it will become DEADLOCK condition.

Task 05:

Explain the code

- Initializes the 2 threads for producer and consumer each one.
- Each producer will produce 3 values, and each consumer will consume 3 values.
- And each time before entering value inside buffer, it checks whether the buffer is already full. If it is full, the producer will get blocked and if it is not full, it will add value inside buffer after locking with mutex and increment the FULL SEMAPHORE. And unlock the mutex.
- Next, each time the consumer consumes the value it will first check whether the buffer is empty. If it is empty, the consumer will get blocked and if it is not empty, it will consume value inside the buffer after locking with mutex and increment the EMPTY SEMAPHORE and unlock the mutex.